

Formats & conventions

The same data can look professional and instantly readable — or messy and confusing. The difference is rarely the numbers. It's the rules you apply consistently : how you name things behind the scenes, and how you use colour, fonts, images and icons out front.

What this key knowledge covers

This summary distils everything you need for **KK17 — Formats & conventions** in Data Analysis. Work through the explanations, then use the worked good-vs-weak answers to see exactly how marks are won and lost. Tick off the revision checklist at the end before a SAC or exam.

Study summary

The scope, in plain words

In the VCAA Study Design, this is the final key-knowledge point of Unit 1 Outcome 1. It reads as **formats and conventions applied to improve the effectiveness of solutions**, and it splits cleanly into two halves:

Naming conventions

Consistent, meaningful names for the parts of your **databases** (tables, fields, queries) and **spreadsheets** (sheets, ranges, columns). Nobody in the audience sees these — but they make the solution easier to build, read and maintain.

Colour · fonts · images · icons

The visual conventions the **audience** does see. Used consistently, they guide the eye, signal meaning and make the visualisation look credible. Used randomly, they fight the message.

A **convention** is just a rule you choose *once* and then apply *every time*. Effectiveness comes from the consistency, not the specific choice. Mia could pick blue for fish or green for fish — what matters is that fish are *always the same colour* on every chart.

Think of a sports club's gear shed

Imagine the storeroom at a footy club. If every jumper is labelled in the same way — Senior_Home_07, Senior_Away_07 — anyone can grab the right one in seconds. If they're labelled jumper, the blue one, #7??, you waste the whole afternoon. That's **naming conventions**.

Now picture the club running onto the ground. Matching guernseys, one logo, club colours everywhere: the crowd reads "that's one team" instantly. If half wore random colours and three

different logos, you couldn't tell who was who. That's **colour, fonts and icons** — visual identity that lets an audience read meaning at a glance.

Exam-ready link: conventions reduce the *effort* the user spends decoding your solution. Low decoding effort = high effectiveness. That sentence works in almost any KK17 answer.

Naming databases & spreadsheets

A naming convention is a shared rule for what you call things. The audience never sees a field name — but consistent naming makes a solution faster to build, easier to read, and far easier to fix later. Mia's golden rules:

- **Be descriptive, not generic.** sightingDate tells you what it holds; Field3 and Sheet1 tell you nothing.
- **Pick one style and stick to it.** Either camelCase (speciesName) or snake_case (species_name) — just never both in one solution.
- **No spaces or odd symbols** in field, sheet or range names. They break formulas and queries. Use an underscore or a capital instead.
- **Prefix by object type** so things sort together: tbl for tables, qry for queries, frm for forms in a database.
- **Stay consistent with case and units** — if one field is lengthCm, don't make the next one WidthMM.

Watch Mia's reef database clean itself up. Toggle the convention on and off:

Colour that means something

Colour is the fastest signal a visualisation has. Used well it **codes categories**, draws the eye to what matters and reinforces a brand. Used badly it becomes noise. The conventions:

- **Consistent colour-coding:** one category, one colour, everywhere. If fish are blue on the bar chart, they're blue on the pie chart and the map too.
- **Limited palette:** a handful of colours, not a rainbow. More colours = more for the eye to decode.
- **Enough contrast:** text must stand off its background, and adjacent bars must be distinguishable — including for colour-blind viewers (avoid red/green as the *only* difference).
- **Purposeful accent:** use one stand-out colour to highlight the key finding, not for decoration.

Fonts: hierarchy, not variety

Beginners reach for lots of fonts to look "creative". Designers do the opposite. The convention is **restraint plus hierarchy**:

- **One or two typefaces, maximum.** One for headings, one for body is plenty.
- **Build a clear hierarchy** with size and weight: big bold title → medium subhead → smaller body. The reader's eye should know the order instantly.
- **Readable sizes & consistent alignment.** Body text large enough to read across a room if it's a poster; everything aligned to the same edges.
- **Match tone to audience** — a clean sans-serif suits a conservation report; a novelty font undermines its credibility.

✗ Three clashing fonts, no hierarchy

✓ One family, clear hierarchy

Images & icons: one visual language

Images set context; icons compress an idea into a glance. Both follow the same convention — **consistency of style and meaning**:

- **One icon set / one style.** All line icons, or all filled — not a mix. A jumbled set looks unprofessional and slows reading.
- **Each icon means one thing.** If a fish icon stands for "fish sightings", don't reuse it elsewhere for "ocean".
- **Consistent size & placement.** Icons the same size, aligned the same way; images cropped to the same shape and resolution.
- **Relevant & properly sourced.** Images must support the data and be credited (ties back to KK07 referencing) — decorative clip-art that adds nothing is clutter.

Spot the inconsistency

✗ Three different icon styles in one row — line, filled, heavy. The eye notices the *style* change before the meaning.

✓ One consistent line style, one size, evenly spaced.

Mia's infographic: before & after

Identical data. The only thing that changed is whether formats and conventions were applied. Flip between them:

- **Colour:** one consistent colour per category, instead of random clashing fills.
- **Fonts:** one family with a clear title→body hierarchy, instead of three novelty fonts.
- **Icons/images:** a single line-icon style at one size.
- **Naming (unseen):** the underlying sheets and fields were renamed MonthlyCounts / sightingDate, so the build is maintainable.

Write it down: a style guide

Conventions only help if they're applied *consistently* — and the way professionals guarantee that is by writing a short **style guide** before they build. The VCAA teaching activity for this dot point literally asks students to act as an organisation and agree a shared set of formats and conventions. Mia's looks like this:

- **Naming:** camelCase fields; tables prefixed tbl, queries qry ; spreadsheet tabs named by content.
- **Colour:** Fish ● · Coral ● · Inverts ● · Algae ● — used identically on every chart.
- **Fonts:** one sans-serif family; 22 px bold titles, 13 px body.
- **Icons:** single line-icon set, 28 px, one icon per category.

In an exam, "the organisation should agree a style guide / set of conventions and apply it consistently" is a high-value sentence — it shows you understand *why* conventions exist, not just what they are.

Read the verb before you answer

KK17 questions live or die on the command word. Match your response length and depth to it:

Structure an "explain" the examiner can mark

For any "explain how a convention improves effectiveness" question, write three moves: Point → Example → Link. Tap through Mia's model answer:

Your turn — write an "explain" for a **font** or **naming** convention. The detector lights up as your answer hits each move:

VCE-style questions with model responses

Attempt each in your head or on paper first, then reveal the weak answer, the strong answer and the marking guide. Award yourself a mark with the self-mark buttons — your total tracks in the dock.

KK17 cheat sheet

- Descriptive, not Field3 / Sheet1
- One case style — camelCase *or* snake_case
- No spaces / special characters
- Prefix by type: tbl, qry, frm
- Consistent units & case across fields
- One colour per category, everywhere
- Limited palette
- Enough contrast (colour-blind safe)
- One accent for the key finding
- One or two typefaces max
- Clear hierarchy: title → subhead → body
- Readable size, consistent alignment
- Tone matches the audience
- One icon set / one style
- One icon = one meaning
- Consistent size & placement
- Relevant, good resolution, sourced

"A convention is a rule chosen once and applied consistently; the consistency lowers the effort builders and audiences spend on a solution, which is what makes it more effective."

Common traps & misconceptions

Exam trap

▲ **Listing features but never linking to effectiveness.** "Use one font" earns little on an *explain*. "Use one font so the audience isn't distracted and reads the hierarchy faster" earns the mark.

Exam trap

▲ **Confusing conventions with components (KK16).** A chart is a *component*; using the *same colours across all charts* is a *convention*. The question is about consistency rules, not the parts themselves.

Exam trap

▲ **Forgetting Half 1.** Many students only write about colour/fonts. Naming conventions for databases and spreadsheets are explicitly examinable — keep a sentence ready.

Exam trap

▲ **Vague "make it look nice".** Examiners want specific conventions: consistent colour-coding, limited palette, font hierarchy, single icon set, no spaces in field names.

Exam trap

▲ **Ignoring the scenario.** A "justify" answer that never mentions the audience or the solution in the stem is generic. Anchor every convention to *this* case.

How to answer exam questions

Read the **command word** first — it sets how much to write. *State/Identify* wants a word or phrase; *Describe* wants features; *Explain* wants reasons and links; *Justify/Evaluate* wants a judgement backed by evidence. Always pull in the **scenario detail** rather than answering in the abstract. The examples below contrast a weak response with a strong one for the same question.

Q1. Define • 2 marks

The term “naming convention”, and state one example of a naming convention that could be applied to a spreadsheet. ()

✗ A weak answer

Weak: "It's when you name your data properly so it looks good." **Why it's weak:** "name it properly" is circular and doesn't define a *convention* (an agreed, consistent rule). No real spreadsheet example given. ~0–1/2.

✓ A strong answer

Strong: "A naming convention is an agreed, consistent rule for how the elements of a solution are named. For a spreadsheet, one example is naming each worksheet tab by its content — e.g. MonthlyCounts — rather than leaving it as Sheet1." **Why it works:** defines convention as a *consistent agreed rule*, then gives a concrete spreadsheet-specific example. 2/2.

How the marks are awarded

- 1 mark — definition: a consistent/agreed rule for naming elements of a solution.
- 1 mark — one valid spreadsheet example (meaningful sheet/range names, consistent case, no spaces, etc.).

Q2. Identify · 3 marks

Mia's database has the field names below. Identify two problems with these names and rewrite each field using a consistent naming convention. the date seen · SP · No. of fish ()

✗ A weak answer

Weak: "They look messy so I'd change them to better names like date, sp2 and fish." **Why it's weak:** "messy" isn't a named problem; the replacements still aren't descriptive or consistent (sp2, fish). No convention identified.

✓ A strong answer

Strong: "Problem 1 — the names contain **spaces and a full stop** (the date seen, No. of fish), which can break queries and formulas. Problem 2 — they are **inconsistent and non-descriptive** (SP is an abbreviation; the others are long phrases). Rewritten in camelCase: sightingDate, speciesName, countObserved." **Why it works:** two distinct, correctly-named problems + consistent, descriptive, space-free replacements.

How the marks are awarded

- 1 mark — problem 1 (spaces / special characters that break formulas or queries).
- 1 mark — problem 2 (inconsistent style / non-descriptive / abbreviation).
- 1 mark — all three rewritten consistently (one case style, descriptive, no spaces).

Q3. Explain · 4 marks

How two visual conventions — one relating to colour and one relating to fonts — could improve the effectiveness of Mia's AGM infographic. ()

✗ A weak answer

Weak: "She should use nice colours and a good font so it looks professional and people like it." **Why it's weak:** no specific convention (which colour rule? which font rule?), and "looks professional" isn't a clear effectiveness link. Generic.

✓ A strong answer

Strong: " Colour — apply consistent colour-coding so each species category keeps one fixed colour across every chart (fish always blue). This lets the audience learn the code once and read all charts without rechecking the legend, communicating findings faster. Fonts — use a single font family with a clear size/weight hierarchy (bold 22 px title, 13 px body). This guides the eye through the information in order, so the audience grasps the structure at a glance, increasing effectiveness." **Why it works:** each convention is named precisely *and* linked to an effectiveness outcome (faster/accurate reading). Two clean PointLink pairs.

How the marks are awarded

- 1 mark — a specific colour convention named (consistent coding / limited palette / contrast).
- 1 mark — colour linked to effectiveness (faster/clearer reading).
- 1 mark — a specific font convention named (single family / hierarchy / readable size).
- 1 mark — font linked to effectiveness.

Q4. Recommend · 4 marks

A classmate's infographic uses five different fonts, a different colour scheme on each chart, and a mix of cartoon and photographic images. Recommend three changes, referring to formats and conventions, that would improve its effectiveness. ()

✗ A weak answer

Weak: "Use fewer fonts, fix the colours, and don't mix the pictures up." **Why it's weak:** the ideas are right but each is a bare instruction with no convention named and no reason. Reads as a checklist, not a recommendation.

✓ A strong answer

Strong: "1 — Reduce to **one or two fonts** with a consistent hierarchy, so the eye isn't distracted and can follow the reading order. 2 — Apply **one consistent colour scheme** across all charts so a colour means the same category throughout, removing the need to re-decode each chart. 3 — Use a **single, consistent image style** (all photographic or all illustrated) so the infographic reads as one cohesive, credible piece rather than a collage." **Why it works:** three named conventions, each with a clear effectiveness reason tied to the audience's reading effort.

How the marks are awarded

- 1 mark each (×3) for a valid change that *names a convention and gives a reason*: font limit/hierarchy; consistent colour scheme/coding; consistent image/icon style.
- Bare instructions with no convention or no reason score partial only.